Correlation

1. Find the co-efficient of correlation for the data n=10, $\overline{x} = 5.5, \overline{y} = 4, \sum x^2 = 385,$ $\sum y^2 = 192, \sum(x+y)^2 = 947.$
2. Find the co-efficient of correlation between x and y for the following data,

x	1	3	4	6	8	9	11	14
y	1	2	4	4	5	7	8	9

3. Obtain the correlation between population density and death rate.

Population density	200	400	500	700	300
Death rate	12	18	16	21	10

4. Calculate the coefficient of correlation between the import and export values.

Export	10	11	14	14	20	22	16	12	15	13
Import	12	14	15	16	21	26	21	15	16	14

5. Find correlation coefficient between supply and price of from following data.

supply	152	158	169	182	160	166	182
Price	198	178	167	152	180	170	162

6. The following marks have been obtained by a class of students in 2 papers of mathematics

Paper-I	45	55	56	58	60	65	68	70	75	80	85
Paper-II	56	50	48	60	62	64	65	70	74	82	90

Calculate the coefficient of correlation for the above data.

Regression

1. Find the regression lines and estimate y for x= 14.5 and x for y=29.5.

x	10	14	19	26	30	34	39
Y	12	16	18	26	29	35	38

2. Obtain line of regression for following data and estimate y for x=4.5.

x	1	2	3	4	5	6	7	8	9
y	9	8	10	12	11	13	14	16	15

3. Two lines of regression are $5y - 8x + 17 = 0$, 2y-5x + 14 = 0. If $\sigma_y^2 = 16$.

   Find: (i) The mean value of x and y (ii) Correlation co-efficient $r(x, y)$ (iii)σ_x.

4. The two-regression equation of the variables x and y are $x = 19.13 - 0.87y$and $y = 11.64 - 0.50x$Find (i) $\overline{x}$ and $\overline{y}$ (ii) The Correlation Coefficient between x and y.

5. The regression lines are $5x - 6y + 90 = 0$ and 15x-8y-130 $= 0$, the variance of x is 9. Find (i) the mean values of x and y. (ii) The correlation x and y. (iii) The S.D of y.

Curve Fitting

1. Fit a straight line of the form $y = mx + c$ to the following data, by using the method of least squares.

x	0	1	2	3	4	5	6	7
y	-5	-3	-1	1	3	5	7	9

2. For the following values of x and y fit a linear curve of the type $y = ax + b$

x	1.0	3.0	5.0	7.0	9.0
y	1.5	2.8	4.0	4.7	6.0

3. Find the best values of a, b, c assuming that the values of x and y are connected by the relation $y = ax^2 + bx + c$

x	1	2	3	4	5
y	3.38	8.25	16.6	28.5	44

4. Fit a parabola of the form $y = ax^2 + bx + c$ to the following data, by using the method of least squares.

x	1	2	3	4	5	6	7
y	-5	-2	5	16	31	50	73

5. A simply supported beam carries a concentrated load p (kg) its middle point. Corresponding to various of p, the maximum deflection y cm is calculated as:

p	100	120	140	160	180	200
y	0.9	1.1	1.2	1.4	1.6	1.7

Find a law of the form $y = ap + b$ using least square criteria.

6. Values of x and y are tabulated as under

X	1	1.5	2.0	2.5
Y	25	56.2	100	156

Find the law of the form $x = ay^b$ to satisfy the given data by least square criteria.

7. Fit a curve $y = ax^b$ using the following data and find values of a and b.

X	2000	3000	4000	5000	6000
Y	15	15.5	16	17	18

8. If the relation between x and y is of the form $y = ab^x$. Using following values of x and y, find the values of constants a and b for the best fitting curve.

x	2.1	2.5	3.1	3.5	4.1
y	5.14	6.788	10.29	13.58	20.578

9. If x and y are connected by the relation $ax^2 + by^2 = x$, find the values of a and b by rearranging the relation into linear form by using least square criteria for following data.

x	1	2	3	4	5
y	3.35	5.92	8.43	10.93	13.45